Chapter 3B — GPU Architecture Roadmap

> *"The AI accelerator market is not a race between two companies. It is a race between two paradigms: NVIDIA's closed, fully-optimized stack versus an emerging open ecosystem. The winner shapes how AI is computed for the next decade."*

Chapter Overview

This chapter traces three generations of AI accelerators — NVIDIA (Hopper → Blackwell → Blackwell Ultra) and AMD (CDNA 3 → CDNA 3+ → CDNA 4) — with the analytical lens of a performance architect rather than a product reviewer. For each generation you understand what architectural problem it solved, what performance it achieves, how to fit it into a hardware selection decision, and where each vendor's roadmap is heading.

Sections in this chapter:

- 3B.1 How to Evaluate a GPU Generation — The Four Lenses
- 3B.2 NVIDIA Hopper — H100 and H200
- 3B.3 NVIDIA Blackwell — B200, GB200, and NVL72
- 3B.4 NVIDIA Rubin — The Next Generation
- 3B.5 AMD CDNA 3 — MI300X Architecture (Corrected Die Count)
- 3B.6 AMD CDNA 3+ and CDNA 4 — MI325X, MI350X, MI355X
- 3B.7 Hardware Selection Framework
- 3B.8 Key Takeaways

3B.1 How to Evaluate a GPU Generation — The Four Lenses

Before diving into specific chips, establish the mental model for evaluating any GPU generation analytically rather than reacting to marketing numbers.

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FOUR LENSES FOR EVALUATING ANY GPU GENERATION:

LENS 1 — COMPUTE DENSITY:
  Metric: Peak TFLOPS (BF16, FP8, FP4), TDP, TFLOPS/W
  Question: How much math can this chip do per watt and per dollar?
  Trap: Peak TFLOPS require aligned workloads. Real workloads achieve 40-70%.
  Better question: What is the sustained TFLOPS for MY matrix shapes?

LENS 2 — MEMORY ARCHITECTURE:
  Metric: HBM capacity (GB), bandwidth (TB/s), $/GB
  Question: How much model fits, how fast can I load weights?
  For inference: HBM bandwidth is usually the binding constraint.
  For training: HBM capacity determines maximum model size without TP/PP.

LENS 3 — INTERCONNECT:
  Metric: NVLink/Infinity Fabric BW (GB/s), IB integration, topology
  Question: How well does this scale beyond a single GPU?
  A GPU that wins on compute but loses on interconnect is unusable for
  training 100B+ parameter models.

LENS 4 — SOFTWARE STACK:
  Metric: CUDA/ROCm ecosystem, kernel library coverage, framework support
  Question: How long does it take to run my workload productively?
  This is NVIDIA's real moat — not just hardware, but 15 years of kernel
  optimization that runs on top of it.

A GPU that wins on compute but loses on interconnect is useless for training.
A GPU with great specs but poor software costs months of engineering to deploy.
Evaluate all four before making a procurement decision.
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3B.2 NVIDIA Hopper Generation — H100 and H200

3B.2.1 H100 SXM5 — The Inflection Point GPU

The H100 (2022) is the chip that defined the modern AI infrastructure era. When AI compute became a national strategic resource, the H100 was the unit of measurement.

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H100 SXM5 KEY SPECIFICATIONS: [SHIPPED]

  Architecture:    Hopper (GH100 die, 80B transistors, TSMC 4nm)
  SM count:        132 SMs
  Tensor Cores:    4th generation — FP8, BF16, TF32, INT8
  Peak BF16:       989 TFLOPS
  Peak FP8:        1,979 TFLOPS  (Transformer Engine enabled)
  HBM3:            80 GB
  HBM3 bandwidth:  3.35 TB/s
  Ridge point:     ~295 FLOP/byte (BF16, dense)
  NVLink 4.0:      900 GB/s bidirectional per GPU
  NVSwitch 3.0:    Full bisection bandwidth in 8-GPU DGX
  TDP:             700W (SXM5)  |  350W (PCIe)
  Process:         TSMC 4nm

CRITICAL HOPPER INNOVATIONS (in order of LLM impact):

1. TRANSFORMER ENGINE (most important):
   Automatically manages FP8/BF16 mixed precision per transformer layer.
   Monitors activation ranges, casts to FP8 for GEMM, accumulates in FP32.
   Result: ~1.9× GEMM throughput vs pure BF16. Zero code changes required.
   This alone makes H100 2× more efficient for LLM training than A100.

2. FP8 TENSOR CORES (E4M3 + E5M2 format pair):
   E4M3: weights and activations (narrow range, high precision)
   E5M2: gradients (wide range for backprop spikes)
   Peak FP8 = 1,979 TFLOPS = exactly 2× BF16 (halved bit width, same tensor count)

3. WGMMA (WARPGROUP MATRIX MULTIPLY-ACCUMULATE):
   Asynchronous MMA instruction: SM issues WGMMA and immediately proceeds.
   Tensor Cores execute concurrently with SMEM prefetch of next tile.
   Enables pipelining of compute and data movement — key to 989 TFLOPS sustain.

4. THREAD BLOCK CLUSTERS:
   Groups of thread blocks that share distributed SMEM across SMs.
   Enables larger effective SMEM pool for giant GEMM tiles.
   Required to hit peak TFLOPS on matrix operations at training scale.

5. NVLINK 4.0 + NVSWITCH 3.0:
   50% more bandwidth per link vs A100 NVLink 3.0 (900 vs 600 GB/s).
   4× NVSwitch chips per DGX vs 6 in A100 era (better bisection per dollar).

PERFORMANCE vs A100 (H100 BF16 vs A100 BF16):
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  Peak BF16 TFLOPS:      989 vs 312  = 3.2× compute (same precision)
  Peak FP8 TFLOPS:       1979 vs N/A = N/A (A100 has no FP8 hardware)
  HBM bandwidth:         3.35 vs 2.0 = 1.68× memory bandwidth
  LLaMA-2 70B training:  ~3× end-to-end speedup (FP8 + TE + larger tiles)
  LLaMA-2 70B inference: ~2.5× end-to-end speedup (BW + FP8)

THE SOFTWARE FLYWHEEL — PERFORMANCE BEYOND LAUNCH SPECS:
  H100 launch (Q4 2022):   989 TFLOPS, ~1,100 tok/s LLaMA-2 70B (8 GPU)
  TRT-LLM (2023):          ~1,900 tok/s (+73%) — continuous batching, fused
kernels
  FP8 inference (2023):    ~2,800 tok/s (+47%) — Transformer Engine inference mode
  FlashAttention-3 (2024): ~3,100 tok/s (+11%) — H100-specific WGMMA optimization
  Cumulative SW gain:      ~2.9× on FIXED silicon over 2 years

  LESSON: Launch specs are not the final performance number.
  NVIDIA's software ecosystem continuously improves hardware utilization.
  Budget 6-12 months from GPU launch to production-quality software.
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3B.2.2 H200 — Memory Refresh, Not Architecture Refresh

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H200 vs H100 — THE CRITICAL DISTINCTION: [SHIPPED]

H200 is NOT a new GPU die.
H200 IS the Hopper GH100 die with upgraded HBM.

  H100 → H200 changes:
    HBM3    → HBM3e    (faster interface standard)
    80 GB   → 141 GB   (+76% capacity)
    3.35 TB/s → 4.8 TB/s (+43% bandwidth)
    700W    → 700W     (identical TDP)
    TFLOPS  → same     (identical Tensor Core count and SM count)

  What this means per workload:
    Compute-bound (training GEMM, large batch prefill):
      H200 = H100 performance  (same Tensor Cores, same compute)
    Memory-bound (decode, small batch inference):
      H200 = 1.43× H100 performance  (proportional to BW improvement)
    Memory-capacity-limited (70B model fitting):
      H200 single GPU vs H100 requires 2 GPUs:
        H100: 80 GB < 140 GB → TP=2, 2 GPUs, AllReduce overhead
        H200: 141 GB ≥ 140 GB → 1 GPU, no AllReduce
        Result: ~40% more throughput on 70B despite identical compute
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  INTERVIEW TRAP:
    "H200 has more TFLOPS than H100" — FALSE. Same compute die.
    "H200 is faster at everything" — FALSE. Equal compute, faster memory-bound.
    "H200 is just a minor refresh" — FALSE. For 70B inference: 1 GPU vs 2 is huge.

STRATEGIC SIGNIFICANCE:
  H200 uses SXM5 form factor — same physical module as H100.
  Existing DGX H100 servers can be upgraded to H200 by swapping GPU modules.
  No rack redesign, no new cooling infrastructure, no cabling changes.
  This is NVIDIA's "server platform continuity" play — upsell customers already
  invested in H100 infrastructure without requiring full cluster replacement.
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3B.3 NVIDIA Blackwell Generation — B200, GB200, and NVL72

3B.3.1 B200 — The Dual-Die Breakthrough

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BLACKWELL B200 ARCHITECTURE: [SHIPPED]

THE FUNDAMENTAL DESIGN CHANGE — Dual-Die MCM:

  Problem: Reticle limit.
    Modern EUV lithography has max exposure area ~858 mm².
    H100 (GH100): 814 mm² — already at the practical reticle limit.
    To add more compute: must use a new process node OR multiple dies.
    NVIDIA chose: Two compute dies + high-bandwidth die-to-die link.

  B200 physical construction:
    2× B100 compute dies (each ~450 mm², TSMC 4NP process)
    Die-to-die connection: NV-HBI (High Bandwidth Interface)
      NV-HBI bandwidth: 10 TB/s  (3× H100's FULL HBM bandwidth)
      NV-HBI latency: effectively zero (on-package, sub-nanosecond)
    Total transistors: 208 billion (vs H100's 80B — 2.6× more)
    HBM3e: 192 GB (2.4× H100's 80 GB)
    HBM bandwidth: 8.0 TB/s (2.4× H100)
    TDP: 1,000W (43% more than H100)

B200 KEY PERFORMANCE NUMBERS: [SHIPPED]
  Peak FP8:        4,500 TFLOPS   (2.3× H100 FP8)
  Peak FP4 (NVFP4): 9,000 TFLOPS (new format, B200-native — 2× FP8)
  HBM3e capacity:  192 GB
  HBM bandwidth:   8.0 TB/s
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  NVLink 5.0:      1,800 GB/s per GPU (2× H100 NVLink 4.0)
  Process:         TSMC 4NP (similar to H100 — not next-gen process)

NV-HBI vs TWO SEPARATE H100s ON NVLINK:
  Two H100s on NVLink: 900 GB/s die-to-die, ~1 μs latency, software-managed
  B200 NV-HBI:         10 TB/s, sub-ns latency, hardware-transparent

  The difference: B200 is a single coherent GPU from software's perspective.
  WGMMA instructions can source tile data from SMEM on either die.
  The programmer writes one kernel; hardware transparently manages die-local data.
  Two separate H100s require explicit TP coordination and AllReduce communication.
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3B.3.2 NVFP4 — The Format That Changes LLM Economics

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NVFP4 (NVIDIA Float4) — THE KEY BLACKWELL INNOVATION:

Format: 4-bit float
  1 sign bit
  2 exponent bits (range: ~0.001 to 448)
  1 mantissa bit
  Block scaling: every 16 weights share one FP8 scale factor

WHY FP4 TRAINING IS A GENUINE ADVANCE:
  FP4 inference: relatively straightforward — quantize weights, lose some
precision.
  FP4 training: requires hardware support for 4-bit gradients with maintained
accuracy.

  Before Blackwell: FP4 training failed because gradient accumulation in 4-bit
  produces catastrophic precision loss and training divergence.

  Blackwell solution:
    NVFP4 format with block-level scaling — each block of 16 weights gets
    its own FP8 scale factor (not just one per tensor).
    This preserves enough dynamic range to maintain gradient accuracy.
    2nd-generation Transformer Engine manages NVFP4/FP8/BF16 per tensor
automatically.

  MLPerf v5.1 result (LLaMA 3.1 405B, 512 GB300 GPUs):
    Achieved target accuracy with FP4 training in 64.6 minutes
    vs estimated ~400 minutes on H100 equivalent
    Speedup: ~6.2× (software optimization + FP4 + architecture)
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CUMULATIVE PRECISION IMPROVEMENT:
  A100 BF16:  312 TFLOPS  (baseline)
  H100 BF16:  989 TFLOPS  (3.2× — new process + larger TC array + wgmma)
  H100 FP8:  1979 TFLOPS  (2× H100 BF16 — halved bit width)
  B200 FP8:  4500 TFLOPS  (2.3× H100 FP8 — bigger chip + optimized TC)
  B200 FP4:  9000 TFLOPS  (2× B200 FP8 — halved bit width again)

  From A100 BF16 to B200 FP4: ~29× in four years.
  (Not all of this is available to every workload — quality constraints apply.)
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3B.3.3 GB200 NVL72 — The Rack as a Single Computer

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GB200 NVL72 SYSTEM: [SHIPPED]

The GB200 "superchip" unit:
  1× Grace CPU (ARM Neoverse V2, 72 Cortex X cores)
  2× B200 GPUs
  NVLink-C2C (CPU↔GPU): 900 GB/s, coherent (7× PCIe Gen5)
  CPU memory: 480 GB LPDDR5X at 960 GB/s (coherently accessible from GPU)
  Form factor: fits in 1U half-rack slot

NVL72 rack configuration:
  36× GB200 superchips
  = 36× Grace CPUs + 72× B200 GPUs

NVLink 5.0 fabric (connecting all 72 B200s):
  1,800 GB/s per B200 GPU
  72 × 1,800 GB/s = 129.6 TB/s aggregate rack NVLink bandwidth

Unified memory:
  72 × 192 GB HBM3e = 13.8 TB GPU memory
  36 × 480 GB LPDDR5X = 17.3 TB CPU memory
  Total: ~30 TB coherently addressable memory pool

Aggregate compute (72 B200s):
  FP8:  72 × 4,500 = 324 PFLOPS
  FP4:  72 × 9,000 = 648 PFLOPS

THE ARCHITECTURAL IMPLICATION:
  LLaMA-3.1 405B in BF16: 810 GB
  NVL72 HBM: 13,800 GB
  405B model fits entirely within NVLink domain with room for 17× model copies.
  TP across all 72 GPUs = NVLink bandwidth (not InfiniBand).
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  No InfiniBand needed for weight matrix AllReduce — NVLink handles it.

  Effective memory bandwidth for KV operations: 72 × 8 TB/s = 576 TB/s
  (aggregate, with NVLink-connected KV distribution across all GPUs)

PUBLISHED BENCHMARK: [SHIPPED]
  LLaMA 3.1 405B offline inference: 3.4× more tokens/sec/GPU vs H200
  Driven by: FP4 compute + NVLink 5.0 bandwidth + unified memory

PRACTICAL IMPLICATION FOR CLUSTER DESIGN:
  A single NVL72 rack replaces what previously required a 32-node InfiniBand
cluster.
  For large model inference: the rack IS the serving cluster.
  For smaller models (7B-70B): the rack handles dozens of parallel model
instances.
  TP topology is now "within rack" rather than "within node" — changes sizing
math.
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3B.4 NVIDIA Rubin — Next Generation

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NVIDIA RUBIN: [ANNOUNCED — details subject to change]

Announced: GTC March 2024
Expected: 2025-2026 production availability

Publicly disclosed information:
  Architecture: Rubin (next after Blackwell)
  Process: TSMC 3nm (N3 or N3P — 1 full node jump from Blackwell's 4NP)
  NVLink 6.0: disclosed; exact bandwidth not confirmed
  New HBM: expected HBM4 integration
  Superchip: Grace Hopper successor — Vera Rubin (Vera CPU + Rubin GPU)

Performance projections (not confirmed by NVIDIA):
  Process shrink from 4nm → 3nm: historically 20-30% TFLOPS/W improvement
  With architectural improvements: potential 2× FP8 throughput vs B200

WHAT IS CONFIRMED:
  NVIDIA is on an annual cadence (H100 → B200 → Rubin annually)
  Rubin Ultra (2026): follows Rubin — both planned and publicly stated
  Grace CPU → Vera CPU: ARM-based, new microarchitecture
  NVLink bandwidth will continue to scale (5.0 → 6.0)
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ENGINEERING NOTE:
  3nm process provides significant power efficiency gain.
  If B200 requires 1,000W for 9 PFLOPS FP4,
  Rubin at 3nm could achieve similar FP4 throughput at ~700W,
  or 2× FP4 at 1,000W — with same cooling infrastructure.
  This is why the annual GPU cadence is credible: process node improvements
  are doing much of the work.
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3B.5 AMD CDNA 3 — MI300X Architecture

3B.5.1 The Correct Die Count (Fixing the Common Misconception)

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MI300X ARCHITECTURE — CORRECTED: [SHIPPED]

⚠️  COMMON ERROR IN THE FIELD: "MI300X is a 3-chiplet design" or "4 compute dies"
✅  CORRECT: MI300X has 10 dies total

Die composition:
  6× XCDs (eXtended Compute Dies) — compute chiplets, TSMC 5nm
    Each XCD: ~50 Compute Units (CDNA 3 Matrix Cores per CU)
    Total: 6 × ~50 = ~304 CUs (some AMD docs cite 304 CUs total)
    Each CU: 4 Matrix Core units (AMD's Tensor Core equivalent)
    Matrix Core: supports BF16, FP16, FP8, INT8, FP64

  4× MCDs (Memory Cache Dies) — memory interface chiplets, TSMC 6nm
    Each MCD: HBM3 memory controllers + HBM3 PHY
    Each MCD: 8 MB Infinity Cache (L3 for that memory partition)
    Each MCD: connects to 3× HBM3 stacks
    4 MCDs × 3 HBM3 stacks = 12 HBM3 stacks total
    12 × 16 GB per stack = 192 GB total HBM3

PACKAGE LAYOUT:
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  │              MI300X OAM Module                   │
  │                                                  │
  │  [XCD 1] [XCD 2] [XCD 3]  ← top layer: 3 XCDs |
  │  [XCD 4] [XCD 5] [XCD 6]  ← second layer       |
  │                                                  |
  |  [MCD 1][MCD 2][MCD 3][MCD 4] ← base layer     |
  |   +HBM3  +HBM3  +HBM3  +HBM3                   |
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  XCDs stacked on MCDs via Through-Silicon Vias (TSVs)
  XCD-to-XCD: Infinity Fabric over TSVs (~900 GB/s)
  MCD-to-XCD: direct stacked connection

WHY "3 CHIPLETS" IS WRONG:
  This error originated from AMD's early block diagrams showing 3 compute
"clusters"
  per XCD layer. Each "cluster" is one half of an XCD. AMD later clarified the
  correct count. For interviews: always say "6 XCDs + 4 MCDs = 10 dies total."

MI300X COMPLETE SPECS: [SHIPPED]
  Total CUs:       304
  Peak BF16:       ~1,307 TFLOPS
  HBM3:            192 GB
  HBM bandwidth:   5.3 TB/s
  Ridge point:     ~247 FLOP/byte
  Infinity Fabric: ~896 GB/s GPU-GPU (OAM-to-OAM, intra-node)
  TDP:             750W
  Form factor:     OAM (Open Accelerator Module)
  L3 cache:        32 MB (4 × 8 MB Infinity Cache on MCDs)
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3B.5.2 Why 192 GB Is AMD's Core Competitive Argument

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THE MEMORY MOAT — QUANTIFIED:

  Model          | BF16 Size | H100 GPUs | MI300X GPUs | Advantage
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  LLaMA-3 8B     |  16 GB    |    1×     |      1×     | Tied
  LLaMA-3 70B    | 140 GB    |    2×     |      1×     | 1 GPU saved ($65K)
  LLaMA-3 405B   | 810 GB    |   12×     |      5×     | 7 GPUs saved ($455K)
  Mixtral 8×22B  | 281 GB    |    4×     |      2×     | 2 GPUs saved ($130K)
  DeepSeek-V3    | ~1.3 TB   |   18×     |      7×     | 11 GPUs saved ($715K)

For 70B BF16 serving — what single-GPU saves:
  Hardware: 1 GPU × $45K AMD vs 2 GPUs × $65K NVIDIA = $45K vs $130K
  Communication: 0 AllReduces vs 160 AllReduces/step (42 ms overhead)
  Operations: 1 process vs 2 processes, 1 network port vs 2
  Reliability: 1 GPU MTBF vs 2 GPU MTBF

WHEN AMD'S ADVANTAGE DISAPPEARS:
  FP8 weights: LLaMA-3 70B FP8 = 70 GB → fits on H100! No AMD advantage.
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  INT4 weights: LLaMA-3 70B INT4 = 35 GB → fits on single H100!

  AMD's memory moat erodes as quantization precision drops.
  At FP8, the argument shifts entirely to raw decode throughput (MI300X 5.3 vs
H100 3.35 TB/s = 58% more).
  At INT4, even A100 (80 GB) fits the 70B model.
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3B.6 AMD CDNA 3+ and CDNA 4 — Roadmap

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AMD AI GPU ROADMAP: [SHIPPED / ANNOUNCED]

  GPU       | Arch   | Process  | XCDs | MCDs | HBM        | BW      | BF16
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  MI300X    | CDNA 3 | 5nm/6nm  |  6   |  4   | 192 GB HBM3|5.3 TB/s |1,307T
[SHIPPED]
  MI325X    | CDNA 3+| 5nm/6nm  |  6   |  4   | 256 GB HBM3e|6.0 TB/s|1,307T
[SHIPPED]
  MI350X    | CDNA 4 | 3nm/6nm  |  6   |  2   | 288 GB HBM3e|8.0 TB/s|~3,655T
[ANNOUNCED]
  MI355X    | CDNA 4 | 3nm/6nm  |  6   |  2   | 288 GB HBM3e|8.0 TB/s|~3,655T
[ANNOUNCED]
  MI400     | CDNA 4+| 3nm      |  TBD |  TBD | >288 GB    | >8 TB/s | TBD
[ROADMAP]
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MI325X — HBM3e Refresh:
  Same compute dies as MI300X (identical 6 XCDs)
  HBM3e replaces HBM3: 256 GB (+33% capacity), 6.0 TB/s (+13% BW)
  Same compute peak: 1,307 TFLOPS BF16
  Use case: workloads where memory capacity is the bottleneck, not compute

  At 256 GB: fits LLaMA-3 405B in INT8 on single GPU (~405 GB).
  This was not achievable on MI300X.

MI350X — CDNA 4: The Step-Change:
  3nm TSMC node (same as Rubin XCDs) — first AMD AI GPU on 3nm
  6 XCDs on 3nm (same count, different process)
  Only 2 MCDs (vs 4 on MI300X): achieves 288 GB via denser HBM3e stacks

  FEWER MCDs WITH MORE MEMORY:
    MI300X: 4 MCDs × 48 GB/MCD = 192 GB
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    MI350X: 2 MCDs × 144 GB/MCD = 288 GB
    Each MI350X MCD holds 3× more HBM — enabled by HBM3e density advances.
    Benefit: fewer dies → lower power, lower cost, better cross-die BW efficiency.

  Compute improvement (3nm → estimated):
    3nm vs 5nm: ~40-50% TFLOPS/W improvement
    With new Matrix Core design: ~2.8× vs MI300X overall
    Peak BF16: ~3,655 TFLOPS [ANNOUNCED — verify before reliance]

  FP4/FP6 SUPPORT — AMD's Blackwell Parity:
    MI350X adds FP4 (AMD's equivalent to NVFP4) and FP6 format support.
    Block scaling architecture mirrors Blackwell's approach.
    This is AMD's answer to "Blackwell has 9 PFLOPS FP4."

MI355X — Liquid Cooled Variant:
  Identical silicon to MI350X
  Direct Liquid Cooling enables higher sustained TDP
  Targets cluster deployments with liquid cooling infrastructure
  Expected: similar performance to MI350X at higher sustained clock
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3B.7 Hardware Selection Framework

This framework converts your workload requirements into a platform selection decision.
It is designed to be used in procurement reviews and system design interviews.

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HARDWARE SELECTION DECISION FRAMEWORK:

STEP 1 — DETERMINE PRIMARY WORKLOAD TYPE:
  A) Training large models (70B+)
  B) Training medium models (7B-30B)
  C) Inference serving — latency-sensitive (P99 TTFT < 300ms)
  D) Inference serving — throughput-oriented
  E) Research / experimentation

STEP 2 — DETERMINE MODEL SIZE AND PRECISION:
  Compute model memory requirement:
    BF16: N_params × 2 bytes
    FP8:  N_params × 1 byte
    INT4: N_params × 0.5 bytes

  Compare to GPU HBM capacity:
    If model_bytes > GPU_HBM: TP required → affects interconnect choice
    TP_degree = ceil(model_bytes / GPU_HBM)

STEP 3 — APPLY THE DECISION TABLE:

  Workload                        | Primary metric  | Leading platform (2025)
  ———————————————————————————————————————————————————————————————————————————
  70B+ BF16 training              | FP8 TFLOPS      | H100 SXM5 / B200 SXM
  70B+ BF16 inference, low-lat.   | HBM cap + BW    | MI300X (single-GPU serving)
  70B+ FP8 inference              | HBM BW          | MI300X (5.3 TB/s) / H200
  7B-30B training                 | FP8 TFLOPS/$.   | H100 SXM5 (software
ecosystem)
  7B-30B inference, BF16          | HBM BW/GPU      | H200 (fits + 4.8 TB/s)
  7B-30B inference, INT4          | HBM BW (35 GB)  | H100 or MI300X (both fit
easily)
  405B+ serving                   | HBM cap         | MI300X (5 GPUs vs 12 H100s)
  Research / custom kernels       | CUDA ecosystem  | H100 (best tooling)
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STEP 4 — CALCULATE TCO SCORE:
  Training TCO per token:
    = (GPU_cost / lifetime) / (tokens_per_hour)
    tokens_per_hour = MFU × GPU_count × peak_TFLOPS × 3600 / FLOPS_per_token

  Inference TCO per token:
    = (GPU_cost + power_cost) / lifetime_tokens
    = (GPU_price/MTBF + kW × $/kWh) / (tokens_per_sec × 3600 × days_deployed)
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STEP 5 — SOFTWARE READINESS CHECK:
  For NVIDIA: any framework works (PyTorch, Megatron, vLLM, TRT-LLM)
  For AMD: verify ROCm support for your specific stack
    vLLM ROCm: production quality ✓
    PyTorch ROCm: production quality ✓
    TRT-LLM: NVIDIA-only × (use AMD's equivalent: MIGraphX/MFMA)
    Custom CUDA kernels: need to port to HIP (weeks of work)
    Triton ROCm: mostly supported (some ops need manual fixing)
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3B.7.1 Generation-over-Generation Performance Summary

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GENERATION-OVER-GENERATION: KEY METRICS [SHIPPED/ANNOUNCED]

  GPU          | Process | BF16 TFLOPS | FP8 TFLOPS | HBM      | BW      |
NVLink/IF
------------------------------------------------------------------------------
  A100 SXM4    | 7nm     | 312         | N/A        | 80 GB    | 2.0 TB/s| 600 GB/s
[S]
  H100 SXM5    | 4nm     | 989         | 1,979      | 80 GB    | 3.35 TB/s|900 GB/s
[S]
  H200 SXM5    | 4nm     | 989         | 1,979      | 141 GB   | 4.8 TB/s| 900 GB/s
[S]
  B200 SXM     | 4NP     | ~2,250      | 4,500      | 192 GB   | 8.0 TB/s|1,800
GB/s [S]
  MI300X       | 5/6nm   | 1,307       | ~2,614     | 192 GB   | 5.3 TB/s| 896 GB/s
[S]
  MI350X       | 3/6nm   | ~3,655      | ~7,310     | 288 GB   | 8.0 TB/s| TBD
[A]
------------------------------------------------------------------------------
  [S] = SHIPPED  [A] = ANNOUNCED

KEY GENERATIONAL INSIGHTS:

1. H100 → H200: Compute flat, memory +76% capacity, +43% BW.
   Pure memory investment. Teaches: for inference, HBM >> FLOPS.

2. H100 → B200: FLOPS 4.5×, memory 2.4×, BW 2.4×.
   Compute grew faster than memory — B200 is more compute-rich relative to BW.
   B200 ridge point: 4500 TFLOPS / 8.0 TB/s = 563 FLOP/byte.
   (vs H100's ~295 FLOP/byte)
```

```
   More workloads are now memory-bound on B200 vs H100.
   Quantization becomes even more important as ridge point rises.

3. MI300X → MI350X (CDNA 4): Compute 2.8×, memory +50%, BW +51%.
   Balanced improvement. FP4 support added.
   AMD closing the FLOPS gap with NVIDIA (MI350X ~3655 vs B200 ~2250 BF16).
   HBM gap reversal: MI350X 288 GB > B200 192 GB.

4. ANNUAL CADENCE REALITY:
   NVIDIA: H100 (2022) → B200 (2024) → Rubin (2025?) → Rubin Ultra (2026)
   AMD: MI300X (2023) → MI325X (2024) → MI350X (2025) → MI400 (2026?)

   Implication: A cluster procured today (H100) will be 2-3 generations behind
   in 3 years. This drives the "buy vs cloud" decision — opex cloud stays current,
   capex on-prem locks in a generation for 3-5 years.
```

3B.8 Key Takeaways

```
CHAPTER 3B — GENERATION SUMMARY FOR THE PRINCIPAL ARCHITECT:

NVIDIA HOPPER (H100 / H200):
  H100 defined the AI infrastructure era. Transformer Engine is the most
  important innovation — zero code changes for 2× throughput.
  H200 is NOT a new architecture — it is HBM3e on the same GH100 die.
  H200's value is single-GPU 70B serving (fits 140 GB model).
  Know cold: H100: 989T BF16 / 3.35 TB/s / 80 GB / 700W
             H200: 989T BF16 / 4.8 TB/s  / 141 GB / 700W

NVIDIA BLACKWELL (B200 / GB200 NVL72):
  Dual-die MCM: two dies connected at 10 TB/s NV-HBI.
  FP4 (NVFP4): genuine hardware advance, not software quantization.
  B200 ridge point: ~563 FLOP/byte (vs H100's ~295) — more workloads memory-bound.
  GB200 NVL72: the rack as a single coherent computer — 30 TB unified memory.
  Know cold: B200: 4500T FP8 / 9000T FP4 / 8.0 TB/s / 192 GB / 1000W

AMD CDNA 3 (MI300X):
  10 dies total: 6 XCDs + 4 MCDs — NOT "3 chiplets."
  192 GB HBM3 is the competitive moat for 70B-405B inference.
  Single-GPU 70B serving: eliminates TP, AllReduce, second GPU cost.
  Software (ROCm): production quality for standard workloads; CUDA still leads.
  Know cold: MI300X: 1307T BF16 / 5.3 TB/s / 192 GB / 750W
```

```
AMD CDNA 4 (MI350X) [ANNOUNCED]:
  3nm process: ~2.8× compute improvement vs MI300X.
  2 MCDs with denser HBM3e: 288 GB (48 GB more than B200).
  FP4/FP6 support: AMD's answer to Blackwell NVFP4.

HARDWARE SELECTION RULE:
  For 70B+ BF16 training: H100 SXM5 (ecosystem + FP8 + software maturity)
  For 70B+ BF16 serving: MI300X (1 GPU vs 2, eliminates AllReduce)
  For 70B+ FP8 serving: H100 or MI300X (both competitive at FP8)
  For research/custom kernels: always NVIDIA (CUDA ecosystem)

THE THREE-YEAR PROCUREMENT REALITY:
  GPU you buy today is 2-3 generations behind at retirement.
  For on-prem: buy when your workload is > 80% GPU-bound continuously.
  Below that: cloud provides current hardware without depreciation risk.
```

Further Reading

1. *NVIDIA Hopper Architecture Whitepaper* (2022). `www.nvidia.com/content/dam/en-zz/Solutions/Data-Center/h100/pdf/hopper-whitepaper.pdf` Read Sections 2-5 in full. The Transformer Engine and WGMMA sections are essential.
2. *NVIDIA Blackwell Architecture Technical Overview* (2024). `www.nvidia.com/en-us/data-center/technologies/blackwell-architecture/` Focus on: dual-die MCM, NV-HBI bandwidth, FP4 block scaling, NVL72 topology.
3. *AMD CDNA 3 Architecture: The AMD Instinct MI300 Series* (2023). `www.amd.com/content/dam/amd/en/documents/instinct-tech-docs/white-papers/amd-cdna3-white-paper.pdf` Read for: XCD/MCD die count (corrected), Infinity Cache, cross-die bandwidth.
4. *MLPerf Training v4.0 Results* (2024). `mlcommons.org/benchmarks/training` Provides objective cross-vendor comparison at defined accuracy targets. More reliable than vendor-provided benchmark numbers for production planning.
5. *Hot Chips 2024 — NVIDIA Blackwell and AMD MI300X Presentations* Search: "Hot Chips 2024 NVIDIA Blackwell" and "Hot Chips 2024 AMD MI300X" Primary source for die-level architecture details beyond what vendor blogs reveal.

— End of Chapter 3B —

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